

Optimal Path Planning in the Presence of Static Obstacles and Uncertain Dynamic Obstacles

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In this paper a method for augmenting the RRT* algorithm introduced by Karaman & Frazzoli¹ such that it can plan in an uncertain dynamic environment is presented. The innovation is primarily in the development of a grid management system capable of handling multiple obstacle types including uncertain obstacles, such that the probability of a cell being occupied at a certain time can be calculated and passed to any planning system. Notably this grid management system is capable of interfacing with any planner that calculates its own state in relation to time as it plans. This allows for a planning algorithm to avoid future collisions, and avoid regions that a moving obstacle could occupy in the future, all while avoiding static obstacles as any conventional planner.

I. Introduction

The path and motion planning problem is a fundamental problem in robotics and autonomous systems. There has yet to be an extensive amount of work done on the problem of path planning in the presence of dynamic obstacles, let alone Uncertain as well. The development then of a system that can manage all these cases at once is necessary to provide more flexibility for the implementation of planners on autonomous systems to provide a more robust and reliable system. Presented here is an RRT* planner that calculates time passed as it plans in conjunction with a hybrid grid system capable of being fed both certain and uncertain static and dynamic obstacles. The grid system is capable of propagating all obstacle types through time to inform a planner.

II. Problem Setup & Algorithms

III. Conclusion

References

¹Sertac Karaman and Emilio Frazzoli. Sampling-based algorithms for optimal motion planning. *The International Journal of Robotics Research*, 30(7):846–894, 2011.

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